

Weekly Progress Report

Amity School of Engineering & Technology
B. Tech: Program

WPR No: 8	Duration: 7 September 2026 – 13 September 2026
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Weekly Progress Report (WPR) for Minor Project

To be filled by Students	
Students Name	Rohit Yadav, Vishal, Hardik Mehlawat (Group 148)
Roll no.	472 (Rohit), 568 (Vishal), 449 (Hardik)
Enrollment no.	A2305223472 (Rohit), A2305223568 (Vishal), A2305223449 (Hardik)
Project Title finalized, if Yes, give name, if NO, give reason	Yes — Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches
Synopsis submitted	Submitted (on Amizone, with offer letter)
Literature review	Extraction complete for all 48 primary studies; taxonomy of detection approaches finalised; all 74 reference identifiers verified against arXiv/Crossref records and BibTeX file built
Technical & Economical Feasibility	Not started
Bill of Material	N/A (literature review project this semester)
Project Progress Schedule (PERT Chart)	Week-wise schedule prepared; PERT chart pending
Design of critical components	Paper structure fixed (8 sections); Section 3 taxonomy finalised across subsections 3.1–3.5; dataset comparison table and the metrics/search-flow figures fixed
Fabrication work (give %)	N/A
Experimental work (give %)	Not started (planned for Semester 8)
Result and Analysis	Not started
Report writing	In progress — Sections 1–5 drafted (about 6,970 words), dataset comparison table and two figures done; comparative analysis next
Signature of students	
Work done in this week	
Sections 4 and 5 of the review paper were drafted this week, together with the dataset comparison table and the two figures they refer to. Section 4 (datasets and benchmarks) runs to 996 words and cites 26 references: it opens with a subsection recording how the review itself was conducted — six databases, eight query seeds, and the record count at each stage (101 records identified, 26 duplicates removed, 75 screened by title and abstract, 74 included and 1 excluded) — then traces how evaluation data has moved from the GAN-era ForenSynths protocol of training on ProGAN and testing on eleven unseen generators, through the diffusion-era GenImage and ArtiFact collections, to in-the-wild sets such as Chameleon and TWIGMA. Its third subsection records the known problems with this data as the source papers state them: the JPEG-compression and image-size bias in GenImage, whose removal raises cross-generator accuracy by more than eleven points; the shortcut a single-generator dataset such as CIFAKE leaves open; the staleness that follows from fixing a generator list at publication; and the twenty-seven-point drop between benchmark accuracy and accuracy on Chameleon's realistic images. The dataset comparison table sets out 11 datasets across seven columns — dataset, introducing study, year, scale, real-image source, generators covered, and the limitation its authors state — with every cell copied from the extraction records; 15 cells are left blank because the records state no value, rather than being filled with a plausible guess. Section 5 (evaluation metrics and reporting practice) runs to 688 words and cites 11 references, defining accuracy, precision, recall, F1, false positive rate, AUC, average precision and mean average precision as eight numbered equations, and then examining how the surveyed studies actually report — different metric combinations over different test suites, uncalibrated decision thresholds, and averages that can hide a total failure on a single generator. Two figures were produced as SVG, the search-flow diagram for Section 4 and a metrics overview panel showing the confusion matrix with the headline formulas for Section 5; both were checked as well-formed files with selectable text so that they survive export into the report. Sections 1 to 5 are now drafted at roughly 6,970 words, supported by one comparison table and two figures. The master literature comparison table covering all 48 primary studies, and Section 6 (comparative analysis) built from it, are scheduled next.	
To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory ☐	
Performance of students is unsatisfactory ☐	
A warning to be issued to student(s) (Name) _____	
Student was not well (Name) _____	
Date: 16 September 2026	Signature of Guide

Date: 16 September 2026